Remote Anomaly Surveillance And Detection

BS CEN-24

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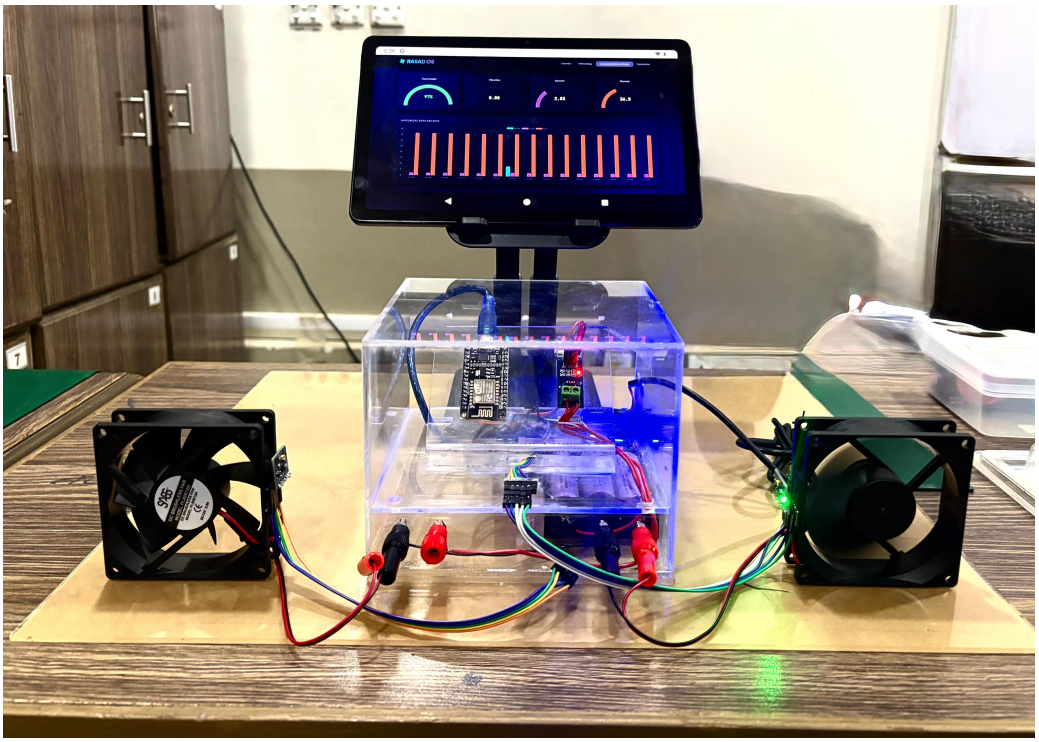


Abstract:

In today's rapidly evolving technological landscape, the integration of Internet of Things (IoT) technologies into appliance health monitoring represents a critical advancement in household safety and energy management. This project — RASAD — presents a smart, sensor-based system designed to continuously monitor the operational health of home appliances in real time.

By leveraging vibration sensors (MPU6050), temperature sensors, and current sensors (ACS712) connected to an ESP8266 microcontroller, the system collects live performance data from appliances such as fans, motors, and electrical loads. This data is processed locally and transmitted to a cloud-based dashboard, enabling users to remotely observe appliance status, detect anomalies, and receive alerts before faults escalate into costly failures.

The system applies predictive maintenance principles to the home environment, identifying early warning signs of appliance degradation — such as abnormal vibration patterns, overheating, or irregular current draw — and notifying users in time to prevent damage. The cloud dashboard allows for historical data analysis, trend visualization, and informed decision-making regarding appliance care and energy usage.

This research demonstrates that intelligent home appliance monitoring is not merely a convenience but a practical necessity for reducing energy wastage, extending appliance lifespan, and improving household safety. RASAD provides a cost-effective, scalable, and user-friendly IoT solution for modern homes.

INTRODUCTION

In today's modern world, home appliances are integral to daily life — from fans and washing machines to refrigerators and air conditioners. However, these appliances are prone to wear, overheating, mechanical faults, and inefficient operation over time. Undetected faults can lead to appliance breakdown, energy wastage, potential fire hazards, and costly replacements.

The **Remote Anomaly Surveillance And Detection** RASAD — is designed to address these challenges by continuously tracking the operational health of household appliances using modern IoT sensor technology. The system monitors three key health indicators: vibration (to detect mechanical faults), temperature (to detect overheating), and current consumption (to detect electrical anomalies).

By placing sensors directly on or near appliances, the system captures real-time data that reflects the true working condition of each device. This data is processed by an ESP8266 microcontroller and uploaded to a cloud platform, where it is analyzed, visualized, and compared against established baseline values. When abnormal readings are detected, the system triggers alerts — enabling users to act before minor issues become major failures.

This project applies the principles of Predictive Maintenance (PdM) to the home environment. While PdM has traditionally been used in industrial settings, RASAD makes this technology accessible and affordable for everyday households. The system helps users monitor appliance health remotely, understand usage patterns, and make data-driven decisions about maintenance and energy efficiency.

History of Predictive Maintenance(PdM)

Predictive Maintenance (PdM) emerged in the 1990s as industrial technologies began to advance. Early maintenance methods relied heavily on fixed schedules and reactive repairs — either replacing components before they failed (often unnecessarily) or waiting for breakdowns to occur. These approaches were costly and inefficient.

As industries sought to reduce operational costs and improve reliability, PdM gained traction by leveraging real-time sensor data and advanced analytics. The integration of sensors allowed for precise monitoring of equipment conditions — measuring vibration, temperature, pressure, and current draw — enabling maintenance teams to identify degradation before failure occurs.

Over time, PdM has become increasingly refined, incorporating AI, machine learning, and cloud connectivity. RASAD brings this same philosophy into the home, applying proven industrial techniques to protect everyday appliances.

What does pdm stand for in maintenance ?

In the context of RASAD, Predictive Maintenance refers to the use of real-time sensor data to predict when a home appliance is likely to develop a fault or fail. By monitoring indicators such as vibration magnitude, operating temperature, and current consumption, the system identifies early warning signs of appliance degradation.

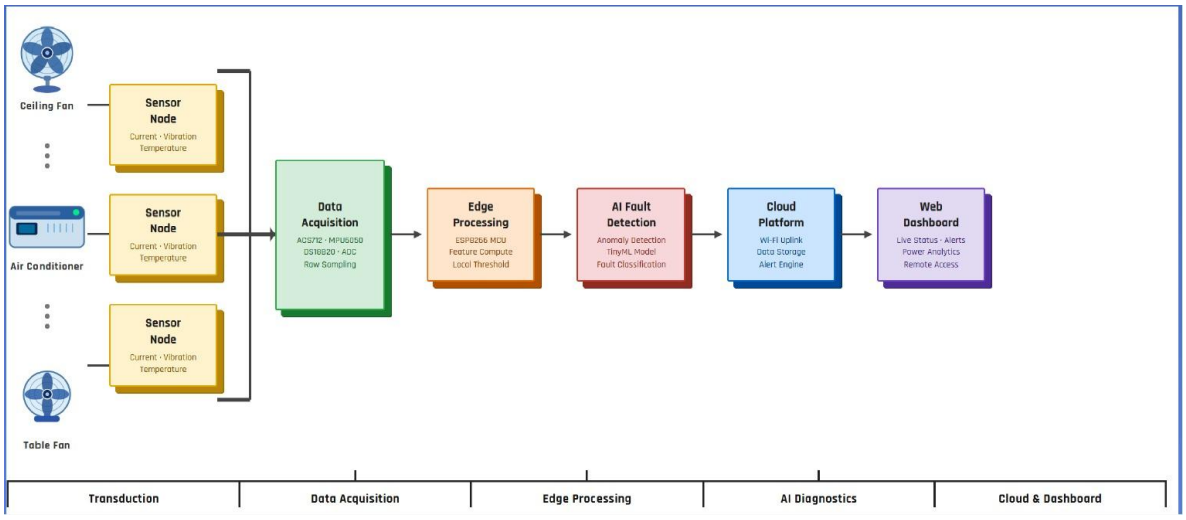
The goal is to reduce unexpected appliance breakdowns, avoid unnecessary maintenance, and extend the operational lifespan of home devices by acting at the right time — based on actual appliance condition rather than guesswork.

OBJECTIVES

The main objectives of the RASAD are:

1. To monitor the operational health of home appliances in real time using IoT sensors.
2. To detect abnormal vibration patterns that indicate mechanical faults or bearing wear in motor-based appliances.
3. To monitor the operating temperature of appliances and detect overheating conditions.
4. To measure electrical current consumption and identify anomalies such as power surges or wire faults.
5. To transmit sensor data to a cloud-based dashboard for remote monitoring and analysis.
6. To generate automated alerts when appliance readings exceed safe thresholds.
7. To help users reduce energy wastage and prevent unexpected appliance failures.
8. To develop a low-cost, reliable, and user-friendly monitoring system using modern IoT technology.

Block Diagram:



METHODOLOGY

The methodology of this project includes the following steps:

1. Study and Analysis

Relevant literature on home appliance faults, IoT sensor technologies, and predictive maintenance strategies was reviewed to establish a solid theoretical foundation.

2. System Planning

A system architecture was designed to integrate multiple sensors with a central microcontroller and cloud platform. The system was planned to support multiple appliances simultaneously.

3. Component Selection

Appropriate sensors were selected based on the health indicators to be monitored: MPU6050 for vibration, a temperature sensor for heat monitoring, and ACS712 for current measurement. The ESP8266 was chosen as the microcontroller for its built-in Wi-Fi capability.

4. System Development

Hardware components were assembled and interfaced with the ESP8266. Firmware was programmed to read sensor values, perform calculations (RMS, magnitude, FFT analysis), and transmit data to the cloud platform via Wi-Fi.

5. Data Collection

The system was deployed on household appliances including fans and motor-based devices. Operational data vibration, temperature, and current was continuously collected during normal and stress-test conditions.

6. Anomaly Detection

Collected data was compared against learned baseline values for each appliance. Threshold-based anomaly detection was implemented to classify readings as Normal, Warning, or Critical.

7. Testing and Verification

The complete system was tested for accuracy, reliability, and response time. Alert notifications were verified under simulated fault conditions.

8. Documentation

All procedures, sensor calculations, results, and observations were documented in this project report.

PROPOSED SYSTEM DESIGN

The proposed system is designed to measure the efficiency and safety of home appliances using modern sensing and communication technologies. The system mainly consists of an **ESP8266 microcontroller**, a **vibration sensor**, a **temperature sensor**, and a **cloud-based monitoring platform**.

The ESP8266 acts as the central processing unit of the system. It collects data from the vibration sensor and temperature sensor, processes the information, and transmits it to the cloud through a Wi-Fi connection. Due to its built-in wireless capability, the ESP8266 provides an efficient and cost-effective solution for real-time data communication.

The vibration sensor is used to detect mechanical movements and vibrations produced by home appliances during operation. These vibrations help in analyzing the working condition and performance of appliances. Abnormal or excessive vibrations may indicate faults, inefficiency, or maintenance requirements.

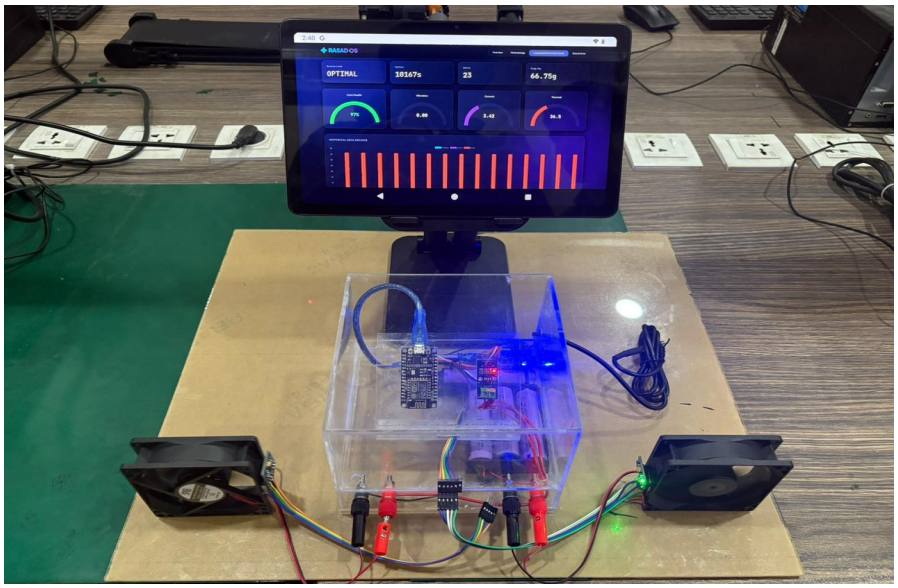
The temperature sensor is used to monitor the operating temperature of appliances. Temperature variations play an important role in determining energy efficiency and safety. Overheating may reduce efficiency and cause damage, while stable temperature indicates proper functioning.

The cloud-based system is used for storing, analyzing, and displaying the collected data. The data sent by the ESP8266 is uploaded to the cloud server, where it can be accessed through a web or mobile interface. This enables users to monitor appliance performance remotely in real time.

The complete system works in a continuous cycle. First, the sensors collect real-time data from appliances. Then, the ESP8266 processes and sends this data to the cloud platform. Finally, the cloud system stores and visualizes the data for analysis and decision-making.

This proposed system provides a reliable, low-cost, and user-friendly solution for monitoring appliance efficiency and safety using IoT technology.

System Design:



SENSORS USED IN THE SYSTEM

1. Vibration Sensor(MPU6050)

Each motor-based appliance has its own MPU6050 mounted directly on it to capture 3-axis acceleration independently.

Note: Bulbs do not require vibration sensors as they have no moving parts.

Calculation Steps

Step 1 Read 3-Axis Acceleration:

Ax, Ay, Az → raw values from MPU6050

Step 2 Calculate Vibration Magnitude:

$V_magnitude = sqrt(Ax^2 + Ay^2 + Az^2)$

Step 3 Remove Gravity Component:

$V_dynamic = V_magnitude - 9.8\ m/s^2$

Step 4 Calculate RMS Vibration:

$V_rms = sqrt((V1^2 + V2^2 + ... + Vn^2) / n)$

Step 5 FFT for Frequency Analysis:

FFT(signal) → Frequency Spectrum

Normal Fan → Dominant peak at 50 Hz

Faulty Bearing → Extra peaks at 120 Hz, 250 Hz

Anomaly Thresholds

WARNING : V_rms > V_baseline × 1.5 (bearing wear starting)

CRITICAL : V_rms > V_baseline × 2.5 (imminent failure)

DIAGNOSE : FFT shows new frequency peaks

2. Temperature Sensor

The temperature sensor monitors the operating temperature of appliances. Temperature is a key factor in determining efficiency and safety. High or fluctuating temperatures may indicate overheating, poor performance, or potential damage, while stable temperatures indicate proper functioning.

3. Current Sensor (ACS712)

Each appliance is fitted with its own CT sensor to measure current independently.

Calculation Steps

Step 1 Convert ADC to Voltage:

V_sensor = (ADC_value / 1023) × V_supply

Step 2 Convert Voltage to Current:

I (A) = (V_sensor - 2.5) / 0.185

Step 3 Calculate RMS Current:

I_rms = I_peak / 1.414

Step 4 Calculate Power:

P (W) = V_supply × I_rms × Power_Factor

Anomaly Thresholds

Each appliance is compared against its own learned baseline:

WARNING : I_current > I_baseline × 1.3 (30% above normal)
CRITICAL : I_current > I_baseline × 1.6 (60% above normal)

INSPECT : I_current < I_baseline × 0.5 (possible wire break)

Cloud Dashboard:

Figure 1 Command and Control Centre

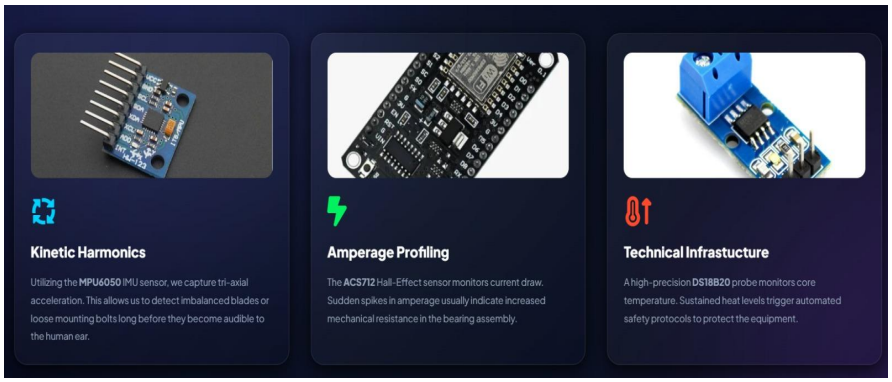


Figure 2 Overview

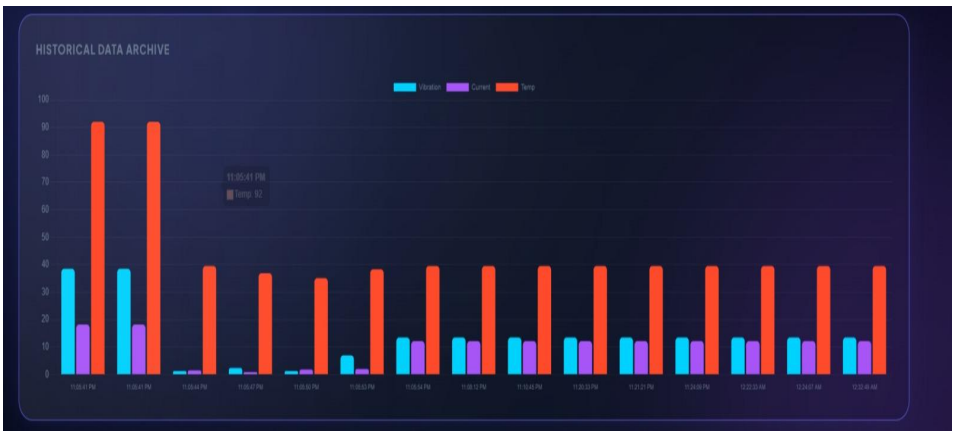


Figure 3 Real time values

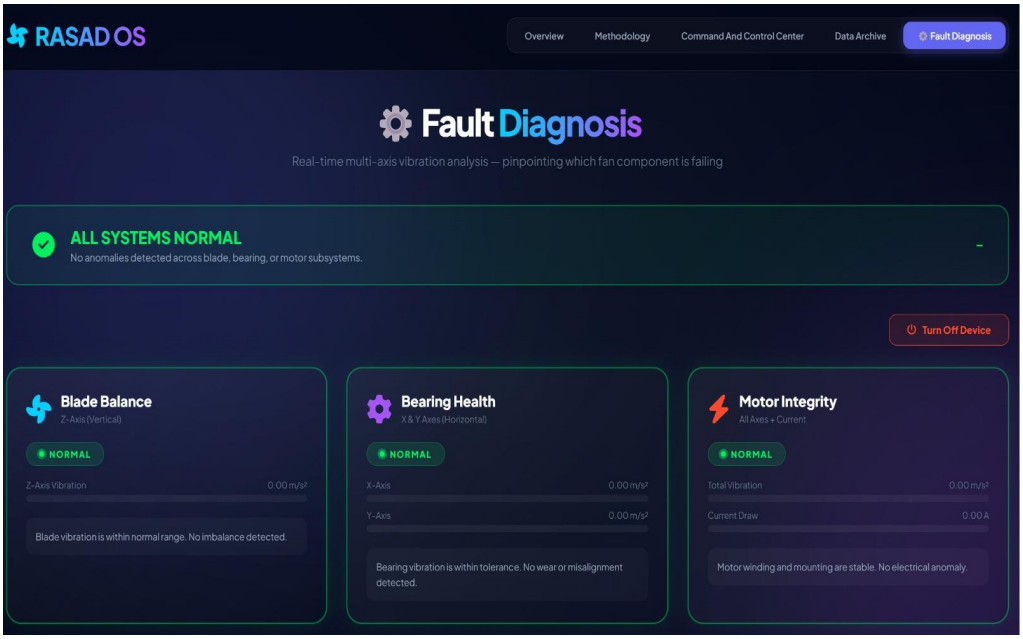


Figure 4 Fault Diagnosis

Scan to Access Live RASAD Dashboard:

Figure 4 Scan to Access Live RASAD Dashboard:

Username:RASAD

Password:1234

Future of Predictive Maintenance

As industries continue to evolve, so too does the field of predictive maintenance. The future promises exciting advancements that will transform how organizations monitor and maintain their equipment. With ongoing innovations in technology, predictive maintenance is poised to become more accurate, efficient, and integral to industrial operations. Here's a look at the key trends shaping the future of predictive maintenance.

1. **Advancements in AI and Machine Learning :**

The future of predictive maintenance will be heavily influenced by advancements in artificial intelligence (AI) and machine learning (ML). These technologies will enable more accurate predictions by analyzing larger datasets and identifying complex patterns. AI and ML algorithms will continue to evolve, improving the precision of predictive models and enhancing decision-making processes.

2. **Integration with IoT :**

The Internet of Things (IoT) will play a crucial role in the future of predictive maintenance. IoT devices will provide real-time data from a wide range of equipment, enabling more granular monitoring and analysis. As IoT technology advances, the integration of IoT sensors with predictive maintenance systems will become more seamless and effective.

3. **Enhanced Data Analytics :**

Future developments in data analytics will drive the evolution of predictive maintenance. Advanced analytics tools will offer deeper insights into equipment health, performance trends, and failure modes. Predictive maintenance solutions(pdms) will leverage big data technologies to process and analyze vast amounts of information, leading to more accurate predictions and optimized maintenance strategies.

4. **Edge Computing :**

Edge computing will enable real-time data processing closer to the source of data collection. This technology will reduce latency and improve the speed of predictive maintenance systems, allowing for faster response times and more immediate decision-making.

5. **Predictive Maintenance as a Service :**

The adoption of predictive maintenance as a service (PMaaS) will grow, offering organizations access to advanced predictive maintenance technologies and expertise without the need for significant upfront investments. PMaaS providers will offer scalable solutions, making it easier for businesses to implement and benefit from predictive maintenance.

6. **Increased Focus on Sustainability :**

The future of predictive maintenance will also include a greater emphasis on sustainability. By optimizing equipment performance and reducing waste, predictive maintenance will contribute to more sustainable operations. Organizations will focus on minimizing environmental impact and promoting energy efficiency through advanced predictive maintenance practices.

CONCLUSION

The **Remote Anomaly Surveillance And Detection** RASAD — successfully demonstrates the application of IoT sensor technology and predictive maintenance principles to the everyday household environment. By integrating the ESP8266 microcontroller with vibration (MPU6050), temperature, and current (ACS712) sensors, the system provides continuous, real-time health monitoring of home appliances.

The system is capable of detecting mechanical faults through vibration analysis, identifying overheating risks through temperature monitoring, and flagging electrical anomalies through current measurement. Anomaly detection thresholds ensure that users receive timely warnings before minor issues escalate into appliance failures or safety hazards.

The cloud-based dashboard enables remote monitoring, historical data analysis, and data-driven decision-making regarding appliance maintenance and energy usage. This makes RASAD a practical and accessible solution for modern households seeking to protect their appliances, reduce energy wastage, and improve overall home safety.

Overall, RASAD represents a cost-effective, scalable, and user-friendly approach to smart home appliance management. It also provided the team with valuable hands-on experience in electronics, embedded programming, IoT systems, and data analysis — skills directly applicable to future engineering endeavors.